

Vision-Based Desktop Automation — Submission Document

Project: Notepad Grounding
Command: `uv run notepad-grounding`
Platform: Windows 11, 1920x1080, 100% scale
Repository: <https://github.com/sofa5060/notepad-grounding>

How It Works

The system uses LLM vision to locate desktop icons without accessibility APIs, window handles, or pre-recorded coordinates. The LLM returns structured output; code owns all coordinate math.

Icon Locating (llm.py)

- Capture a full desktop screenshot (1920x1080)
- Send it to the LLM with a prompt describing the target icon
- LLM returns structured JSON via `text_format` with Pydantic schema validation
- Confidence gating: minimum 0.85 confidence required; up to 3 retries with 1s delays on low confidence or API errors. If all attempts are below threshold, the run is skipped entirely (no click occurs).
- Result includes center point (x, y) and bounding box, both clamped to screen dimensions
- Annotated screenshot saved showing bounding box, crosshair, and confidence

Automation Flow (main.py + automation.py)

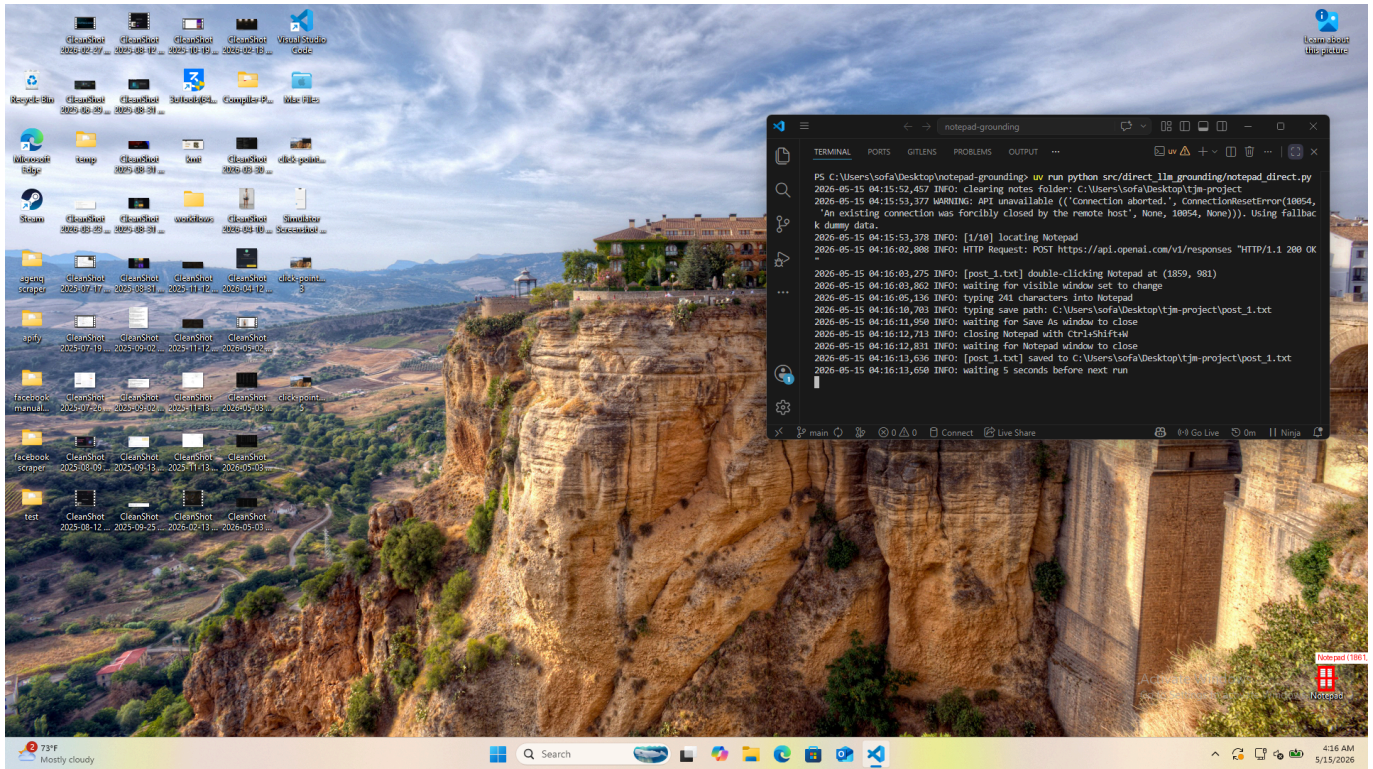
- Take screenshot → locate Notepad icon
- Double-click the icon center
- Wait for window launch (visible window set change detection via Win32 API)
- Type post content (Title + body)
- Save via Ctrl+Shift+S → type path → Enter
- Close via Ctrl+Shift+W (Alt+F4 fallback)
- Repeat for 10 posts

Error Handling

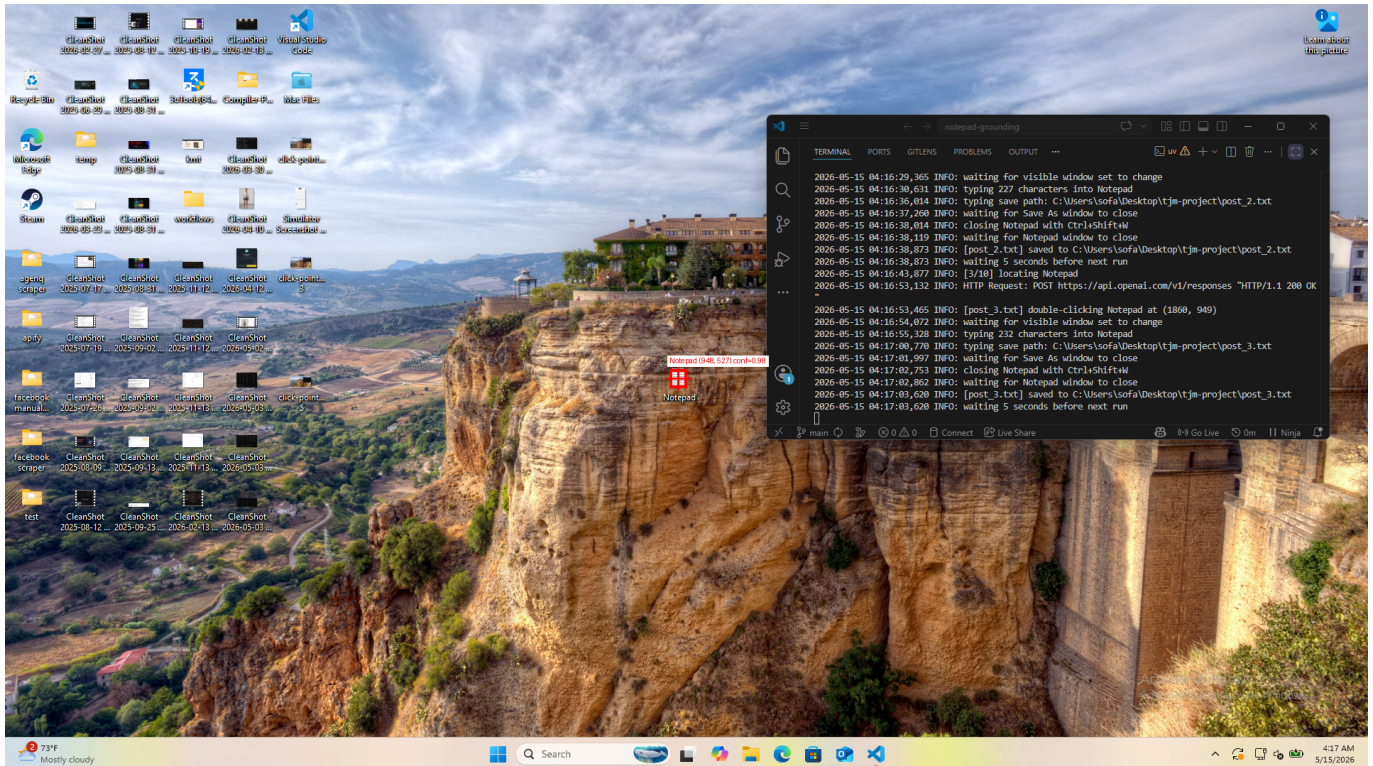
Scenario	Handling
Icon not found / low confidence	Retry up to 3 times, 1s delay; skip run if still below 0.85
API unavailable (fetching posts)	Retry up to 3 times, 5s delay; graceful fallback to dummy data
Window fails to open	Timeout after 8 seconds
Close fails (Ctrl+Shift+W)	Alt+F4 fallback
Existing files in target directory	Directory cleaned before each run

Test Evidence

Case 1: Bottom-Right — Small Icons, Busy Background



Case 2: Center — Small Icons, Busy Background



Case 3: Top-Left — Large Icons, Busy Background



- **3 screen positions:** top-left, center, bottom-right
- **3 icon sizes:** small, medium, large
- **Busy/cluttered backgrounds**

- **Structured output** with confidence gating and retry logic